

Introduction

- Diabetes screening currently depends on finger-prick tests, which are painful, inconvenient, and costly for routine use. This leads many people to skip daily monitoring, increasing the risk of late detection.
- Breath acetone naturally increases when glucose metabolism is impaired, making it a strong non-invasive indicator of diabetes risk. Measuring breath is painless, hygienic, and suitable for frequent checks.
- This project develops a handheld breath analyser using a straw-based sampling system for clean and consistent acetone detection. A multi-sensor array captures real-time breath data for reliable screening.
- A supporting Android application guides users through the test and records the results, enabling trend analysis over time. The system is low-cost, portable, and ideal for homes, clinics, and community health camps.

Motivation

- Routine diabetes checks are difficult for many people, as traditional finger-prick tests involve needles, strips, and proper disposal, making the process painful, costly, and stressful.
- Older adults, children, and frequent testers often skip monitoring due to fear of pricking, recurring expenses, and the hassle of maintaining test supplies.
- A non-invasive breath-based device provides a simple 30–60 second test, using a paper straw and a compact unit that anyone can use before meals, exercise, or bedtime.
- By removing pain, reducing ongoing costs, and improving accessibility, the device becomes ideal for homes, clinics, rural areas, and large community screening camps.

Literature Survey

Article	Approach/Method	Advantages	Limitations	Remarks
A preliminary screening system for diabetes based on in-car electronic nose sensor array (2021)	32-sensor MOS array (TGS2600, TGS2602, TGS2610, TGS2620, MP series) with PCA + SVM classification for breath analysis	Large sensor array provides comprehensive gas coverage; proven classification accuracy; established sensor selection baseline	Bulky system not portable; requires controlled environment; high cost with 32 sensors	Foundational work that established sensor families (Figaro TGS, Winsen MQ/MP) widely used in subsequent studies. Provides evidence-based sensor selection for portable devices.
Hand-held device quantifies breath acetone for real-life metabolic health monitoring (2022)	Compact device using SGP40 sensor with catalytic filter and micro-separation column for selective acetone detection	Truly portable handheld design; selective acetone measurement; real-world usability demonstrated	Single sensor limits cross-validation; requires specialised filter materials; limited to acetone only	Demonstrates feasibility of handheld breath devices. Shows importance of pre-conditioning for selectivity, though filter availability may be challenging for replication.
Mixed potential type acetone sensor with ultralow detection limit for diabetic ketosis breath analysis (2021)	Custom Gd ₂ Zr ₂ O ₇ solid electrolyte sensor with 10 ppb detection limit; tested on healthy vs diabetic breath samples	Extremely low detection limit (10 ppb-100 ppm range); high selectivity; validated with real breath samples	Custom sensor not commercially available; requires high-temperature operation (560°C); complex fabrication	Establishes clear cutoff values (1.8 ppm) for healthy vs diabetic classification. Proves breath acetone detection feasibility but requires custom sensor development.

Literature Survey

Article	Approach/Method	Advantages	Limitations	Remarks
Nanowire Array Breath Acetone Sensor for Diabetes Monitoring (2024)	Self-powered InP nanowire sensor with chitosan coating; covers 0.4 ppb to >100,000 ppm range	Ultra-wide detection range; room temperature operation; integrated into "Ketowhistle" prototype device	Nanowire fabrication complexity; research-stage technology; limited commercial availability	Shows progression toward practical devices with extremely wide range covering healthy (300-800 ppb) to DKA (>75 ppm). Demonstrates prototype integration potential.
Research progress of electronic nose technology in exhaled breath analysis (2022)	Review of various MOS sensor combinations (TGS2600, TGS2602, MQ3, MQ135, MQ137) with ML approaches	Comprehensive sensor technology overview; multiple gas detection; established ML classification methods	Review paper - no specific device implementation; sensor selection varies across studies	Validates common sensor families across multiple studies. Confirms TGS and MQ series as standard choices for breath analysis applications.
Is Breath Best: A Systematic Review on Accuracy and Utility of Nanotechnology Based Breath Analysis (2023)	Systematic review of breath acetone detection methods; reports healthy range 0.3-1.0 ppm	Evidence-based ranges for healthy vs diabetic breath; high correlation with blood ketones (R^2 up to 0.98)	Methodology variations across studies; lack of standardized cutoff values; device comparability issues	Highlights need for standardization in breath testing. Confirms breath acetone as viable biomarker but emphasizes challenges in establishing universal thresholds.

Gaps

- Most existing breath-based diabetes detection systems still use laboratory-style sampling methods, such as Tedlar bags or balloons, which are costly, inconvenient, and prone to user handling errors.
- Current prototypes are often bulky, desktop-bound, or tethered to computers, making them unsuitable for field deployment or daily personal use.
- Many research systems use imbalanced sensor arrays, where too few sensors reduce accuracy and too many increase cost, complexity, and power consumption.
- Existing devices typically present raw sensor waveforms instead of clear screening outcomes, making them difficult for non-technical users, caregivers, or health workers to interpret.
- Environmental variations such as temperature and humidity are rarely compensated, leading to inconsistent readings and limited real-world reliability.

Problem Statement

- To develop a non-invasive, handheld breath-testing device for diabetes screening that eliminates the need for painful finger-prick blood tests. The system will collect breath samples through a simple paper straw, process them using a mixed sensor array, and transmit the data to a mobile application via Bluetooth. The app will guide users through the test, analyse sensor responses, and display clear “Normal” or “Needs Attention” results-providing a quick, comfortable, and low-cost method for routine diabetes screening.

Objectives

Phase 1: Hardware Development

1. **Design and build the non-invasive breath-sampling unit** using a paper-straw intake system with one-way airflow for clean and consistent sampling.
2. **Integrate the mixed sensor array** (MQ sensors, WSP2110, DHT22) with the ESP32 and dual ADS1115 ADC modules for stable multi-channel data acquisition.
3. **Implement reliable firmware functions** for sensor calibration, baseline stabilisation, and real-time data collection during the 30-second test cycle.

Phase 2: Mobile Application Development

4. **Develop the Android app with Bluetooth communication** for connecting, calibrating, and initiating breath tests through guided prompts.
5. **Implement clear result visualisation and user interaction**, providing step-by-step instructions, condition alerts, and test summaries.
6. **Enable secure offline data storage**, saving each trial in timestamped CSV format for trend viewing and easy export when needed.

Objectives

Objective 1:

To design a non-invasive straw-based breath sampling mechanism for clean and consistent acetone detection.

Proposed:

To develop a simple, hygienic breath intake system using a paper straw, incorporating one-way airflow and a guided 30-second sampling protocol to ensure reliable and repeatable breath collection across users.

Submitted to:

Dr. Karthik N, Assistant Professor, Department of CSE, NIT Puducherry

Status:

Completed with a functioning breath-sampling unit that supports baseline stabilisation, guided breath collection, and recovery phases.

Objectives

Objective 2:

To develop a mixed-sensor array and embedded hardware platform for accurate real-time breath analysis.

Proposed:

To integrate MQ-series sensors, an electrochemical acetone sensor (WSP2110), and a DHT22 into a stable hardware system powered by ESP32 and dual ADS1115 modules, enabling synchronized multi-channel data acquisition with proper calibration.

Submitted to:

Dr. Karthik N, Assistant Professor, Department of CSE, NIT Puducherry

Status:

Completed with all sensors connected, calibrated, and capturing reliable voltage readings through the ESP32-based hardware setup.

Objectives

Objective 3:

To develop an Android application for guided breath testing, Bluetooth communication, and result visualisation.

Proposed:

To build a user-friendly mobile app that connects to the breath analyser via Bluetooth, provides calibration and test initiation controls, guides users through each stage of breath sampling, and displays clear screening results along with saving test data for offline access.

Submitted to:

Dr. Karthik N, Assistant Professor, Department of CSE, NIT Puducherry

Status:

Completed and integrated. The Android app now supports Bluetooth communication, guided test workflow, result display, and automatic CSV-based data logging.

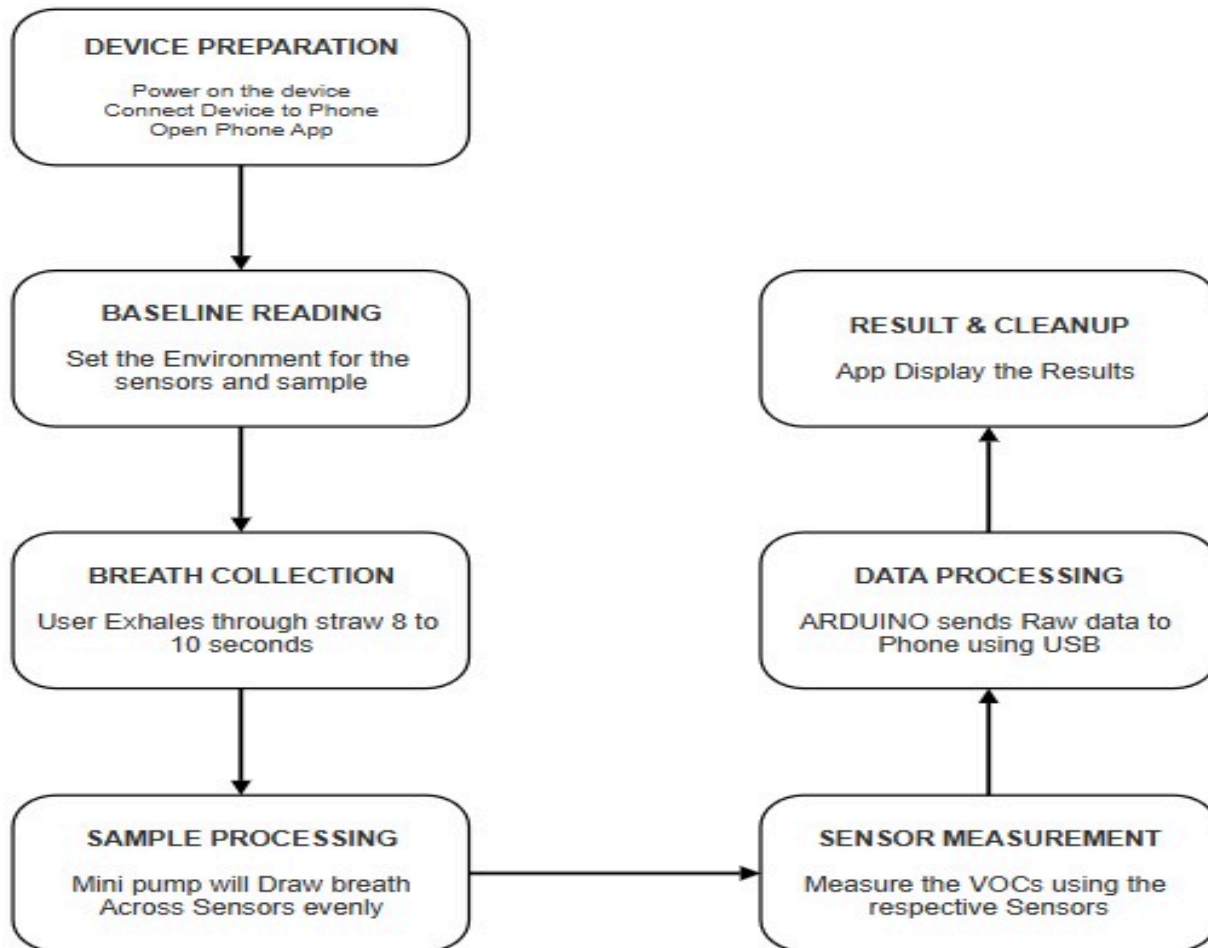
Dataset

For this project, breath samples were collected from a total of **40 participants**, including **20 non-diabetic** individuals and **20 diabetic** individuals. Each participant performed the guided breath test, and sensor readings were recorded across all channels to capture variations in acetone and related volatile compounds. From this dataset, we calculated **sensor-specific thresholds** that differentiate typical diabetic and non-diabetic responses. A **sensor voting system** was then applied: if a majority of sensors crossed their respective thresholds during a breath trial, the result was classified as *diabetic*; otherwise, it was labeled *non-diabetic*. This simple, interpretable method allowed us to validate the device's screening capability without using machine-learning models.

Methodology

- Designed a portable straw-based breath sampling mechanism that collects fresh exhaled air without pumps or Tedlar bags, ensuring clean and user-friendly sample intake. Implemented a guided 30-second sampling cycle (baseline → breath → recovery) with environmental monitoring using temperature and humidity sensors for consistent testing conditions.
- Integrated a mixed multi-sensor array consisting of six MQ-series gas sensors and a WSP2110 electrochemical acetone sensor to capture acetone-specific and cross-sensitivity responses. Constructed stable voltage-divider circuits with decoupling capacitors to ensure noise-free analog readings and sensor stability across trials.
- Utilised dual ADS1115 ADC modules for high-resolution, synchronised digitisation of all sensor channels without signal interference. Employed an ESP32 microcontroller for calibration, timing control, sensor heating cycles, data acquisition, and Bluetooth communication with the mobile app.
- Designed a reliable power subsystem using a 2-cell lithium battery and LM2596 regulators to provide consistent 5V for heaters and 3.3V for logic modules during field operation.
- Developed an Android application that connects via Bluetooth, guides users through each test step, receives real-time sensor data, and stores results in structured CSV format.
- Provided an intuitive app interface that offers clear result visualisation, calibration options, and test history, ensuring accessibility for all types of users. Ensured the entire system operates as a compact, portable, and fully integrated device, suitable for homes, clinics, and large-scale community screening.

Proposed Architecture



Experimental Setup

The experimental setup consists of a complete handheld hardware prototype enclosed in a compact chamber along with a breadboard-based sensor integration layout for development and testing.

A straw-based breath inlet and semi-sealed chamber allow the exhaled air to circulate evenly around the mixed sensor array, reducing external airflow disturbances and ensuring uniform gas exposure.

The sensor module includes six MQ-series sensors and a WSP2110 electrochemical acetone sensor, supported by stable voltage-divider circuits and wiring based on the theoretical reference diagram.

Structural reinforcements using cardboard partitions and hot-glue prevent vibration, loose contacts, and mechanical shifts, ensuring stable sensor readings during repeated trials.

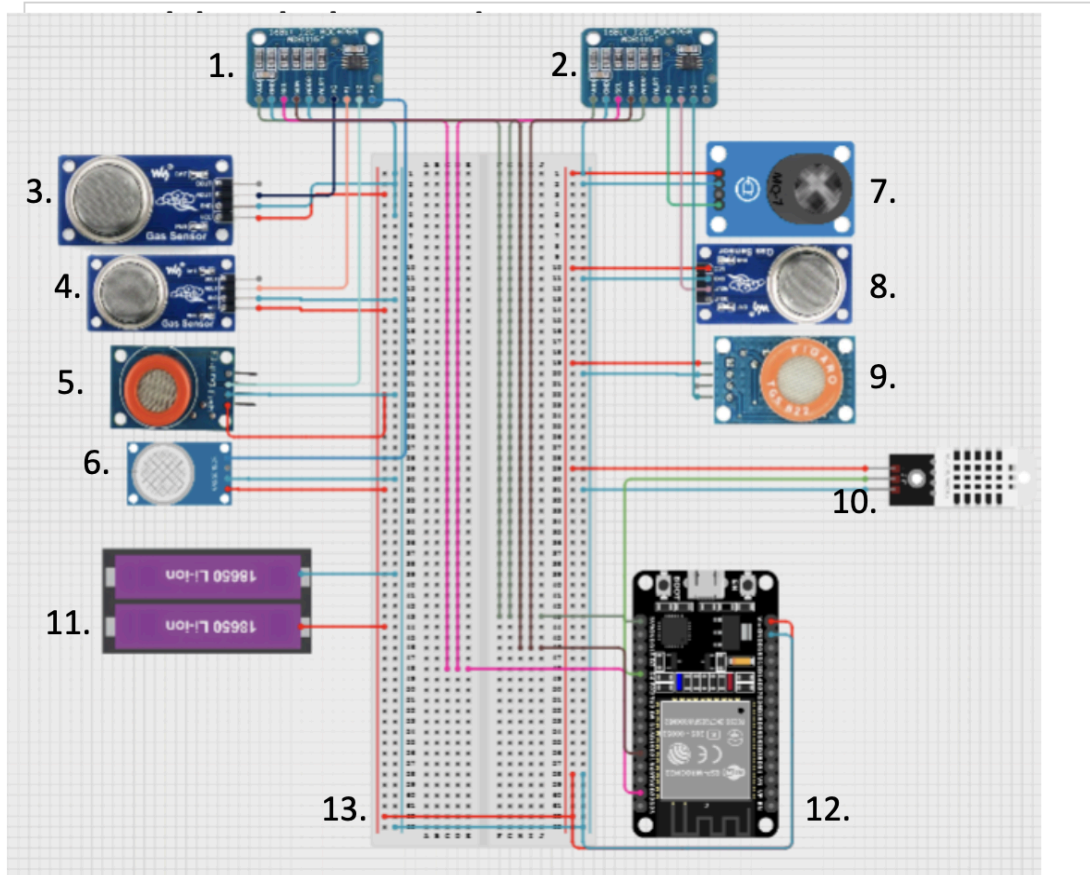
A dual 18650 lithium-ion battery pack with an LM2596 buck converter powers the heaters and logic components, providing the consistent current required for accurate VOC detection.

The ESP32 microcontroller handles data acquisition and wireless communication, transmitting real-time sensor readings to the Android application via Bluetooth.

The setup enables portable, cable-free operation, making it suitable for field testing and for evaluating breath-acetone responses across multiple users.



Implementation



Circuit Diagram:

1 , 2 : ADS1115 used to read analog channels.

3 : MQ-2 Flammable gas/vapor response channel reference.

4 : MQ-3 Alcohol vapor sensitivity for breath.

5 : MQ-4 Methane detection, hydrocarbon reference channel.

6 : MQ-6 LPG/propane response, cross-sensitivity channel reference.

7 : MQ-7 Carbon monoxide response, heater-cycled channel.

8 : MQ-135 Air quality VOC baseline reference.

9 : WSP2110 Electrochemical acetone sensing element, A/B.

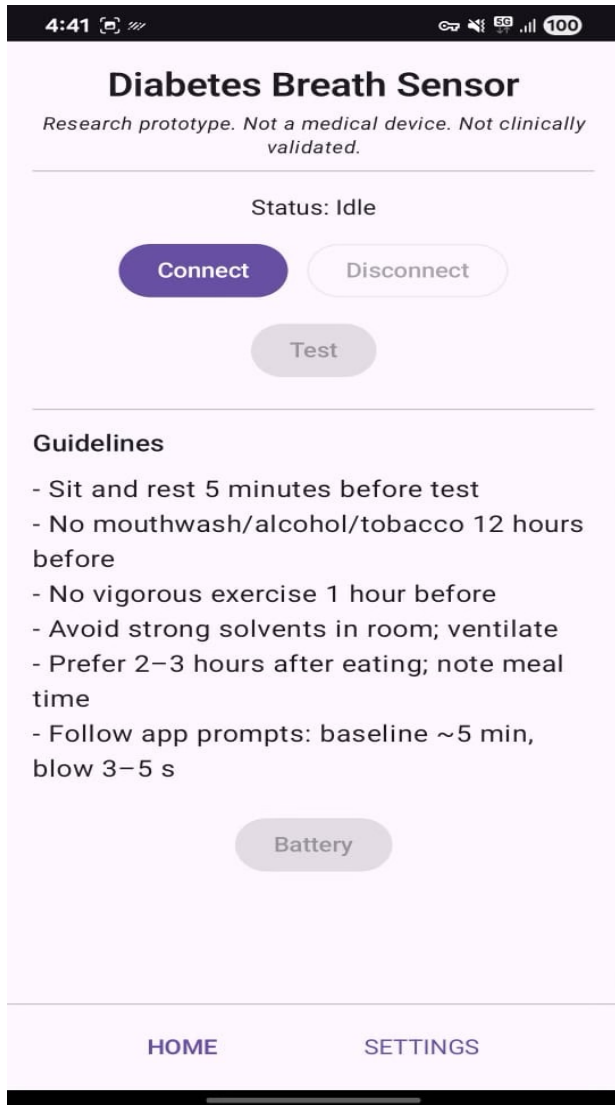
10 : DHT22 Temperature and humidity for compensation.

11 : Battery.

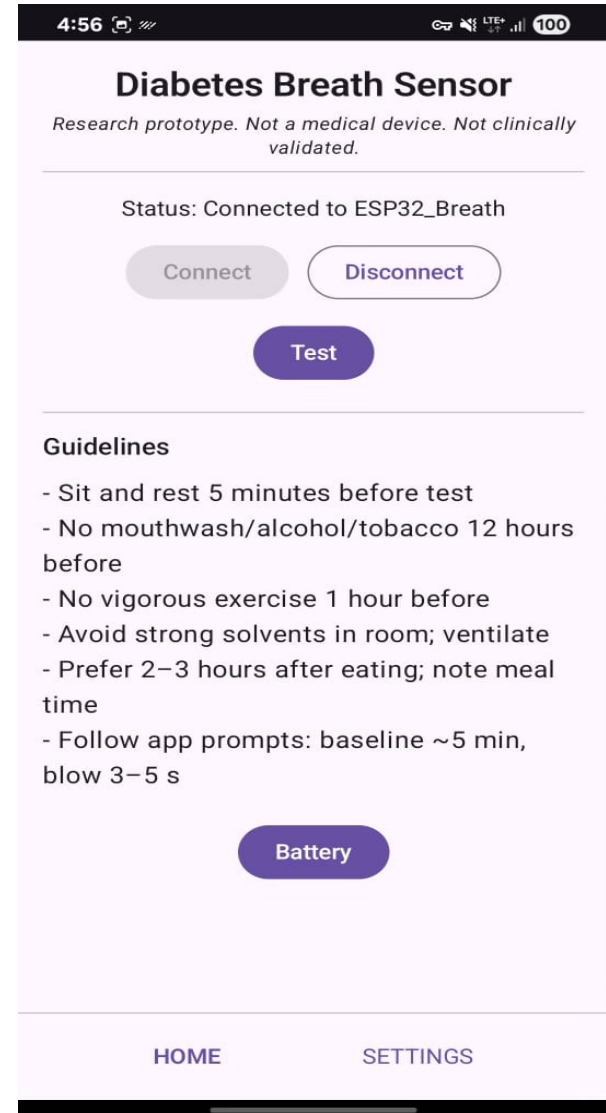
12 : ESP32 DevKit Main controller: Bluetooth, I2C, logic.

13 : Bread Board.

Implementation

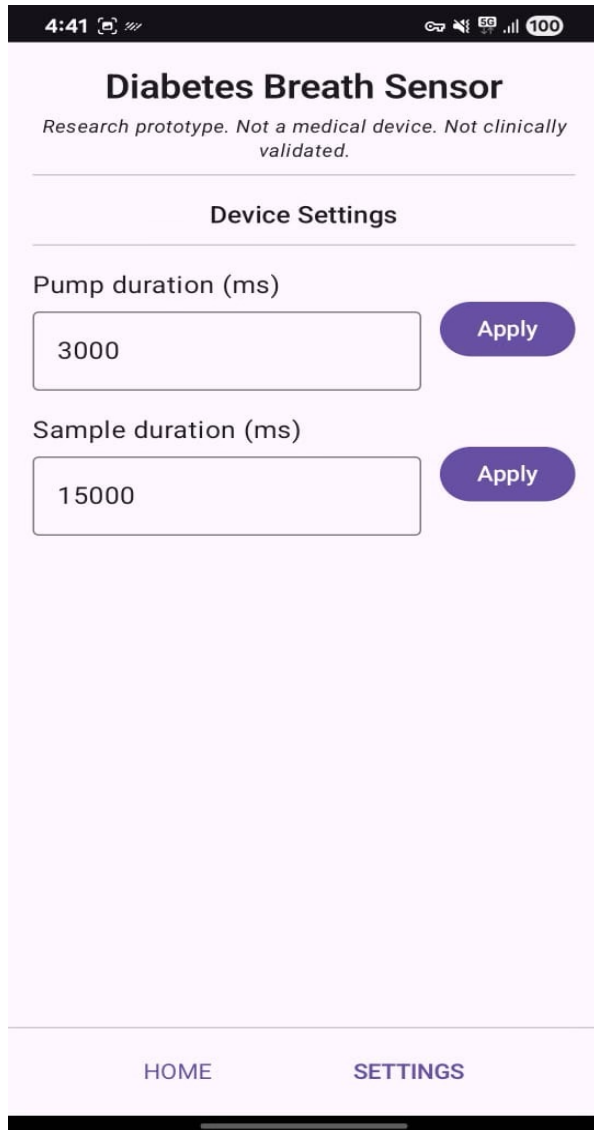


1. Device Not Connected

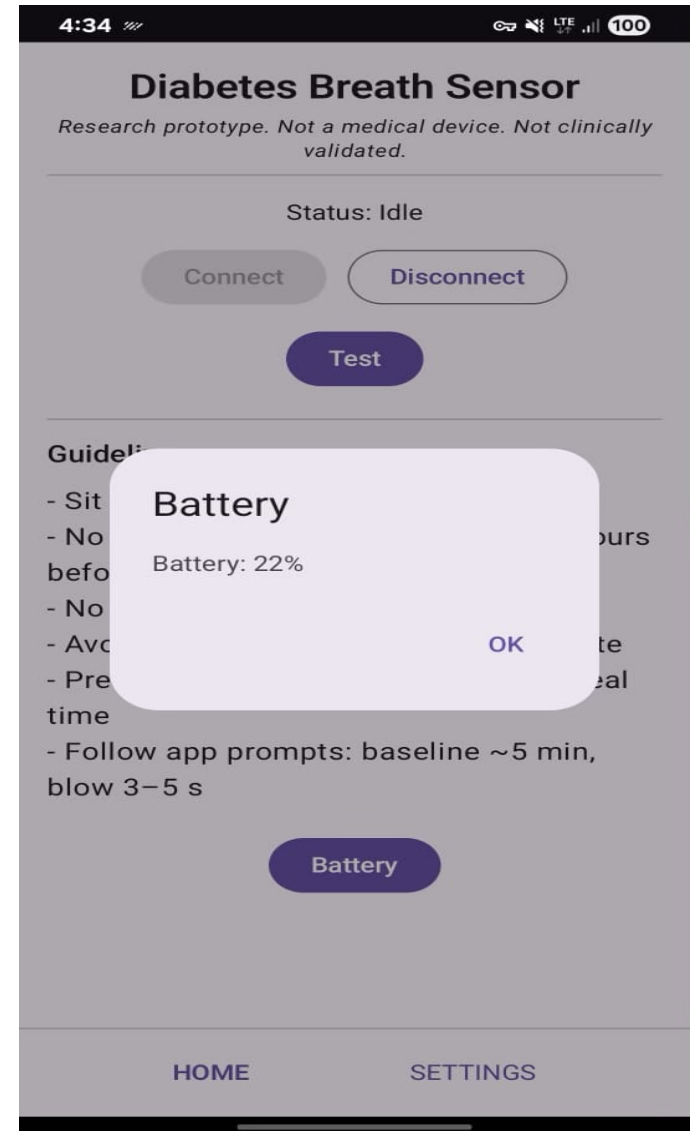


2. Device Connected

Implementation



3.Device Settings Screen



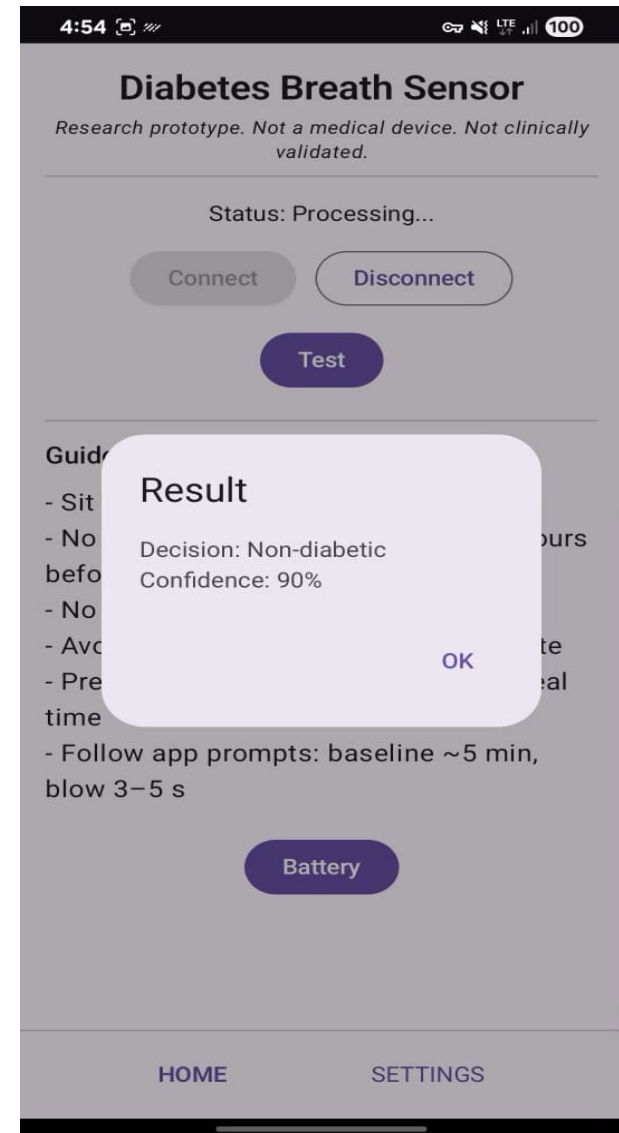
4.Battery Status Popup

Result And Analysis

The device analyses the breath test by comparing the **sample values against the baseline values** recorded during the initial stabilisation phase. Each sensor evaluates how much the sample deviates from the baseline and classifies the response as normal or abnormal. Based on these individual sensor outcomes, the system determines whether the overall breath profile indicates a *diabetic* or *non-diabetic* pattern. A confidence score is then calculated using a voting method, making the result easy to interpret.

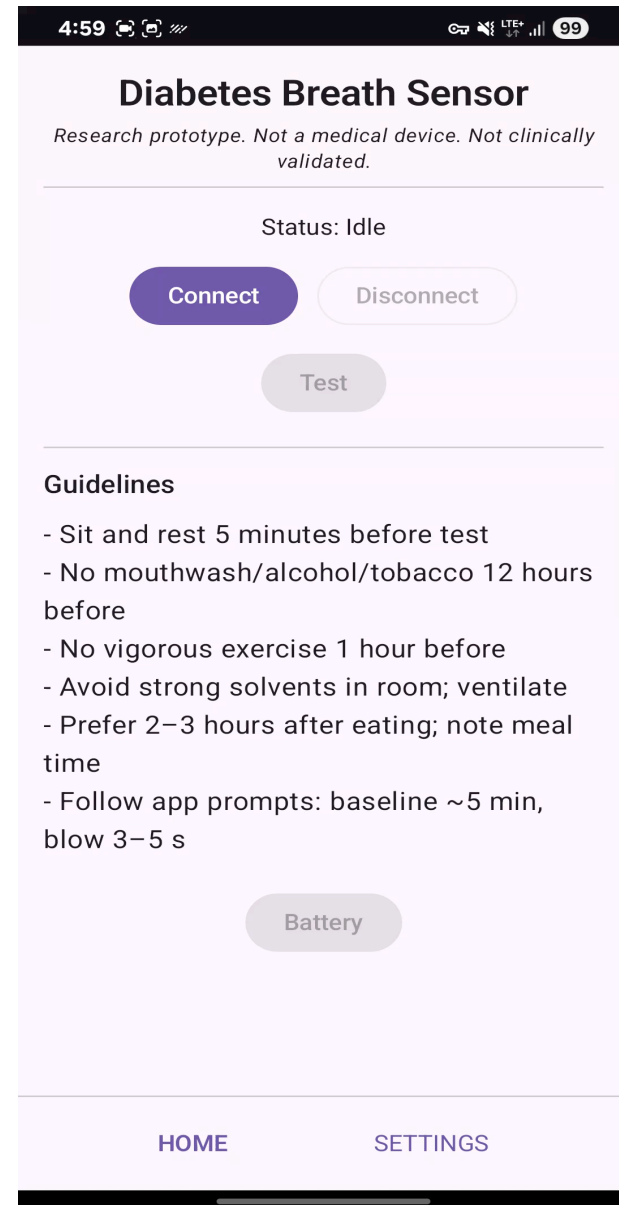
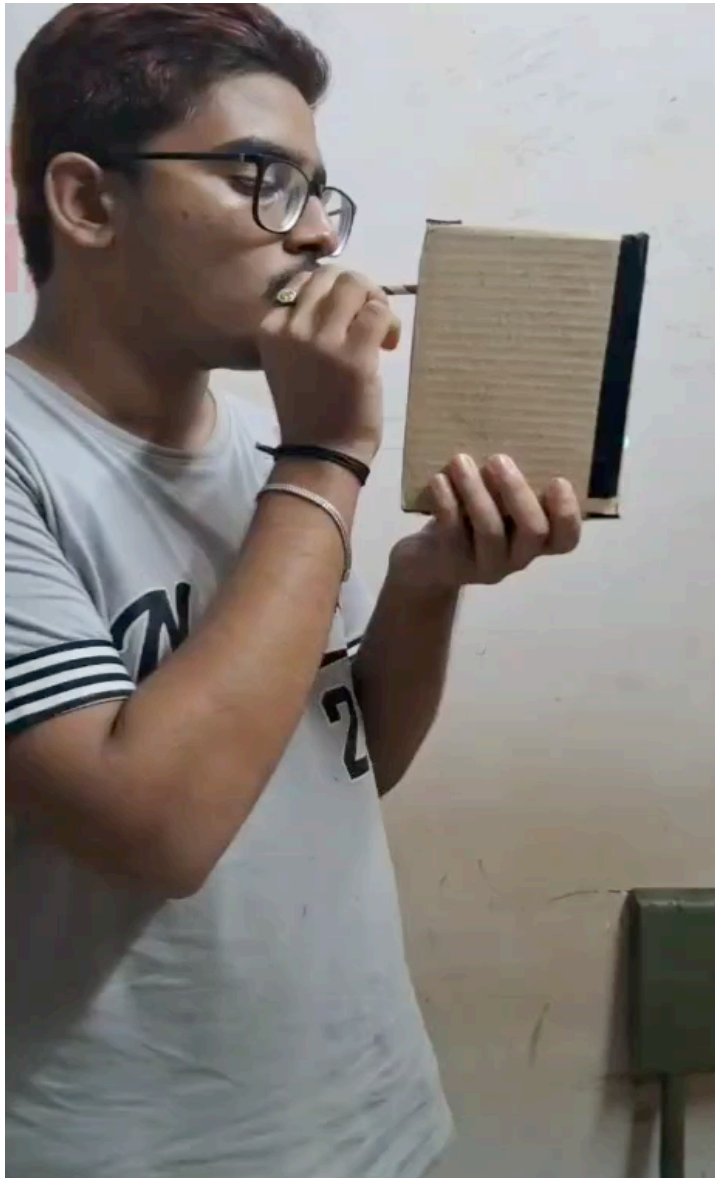
Evaluation:

- **Sample – Baseline** difference is used to determine sensor response.
- **Positive (Abnormal)** = elevated shift → diabetic tendency.
- **Negative (Normal)** = minimal shift → non-diabetic tendency.
- **More normal readings** → **Non-diabetic result**.
- **More abnormal readings** → **Diabetic result**.
- **Confidence level** is based on voting:
 $(\text{number of sensors supporting the result} / \text{total sensors}) \times 100\%$.



5.Result Popup

Live demo of the model



Trade-off of the Proposed Architecture

The proposed non-invasive breath-based screening device offers a low-cost, painless, and portable alternative to traditional blood-based glucose testing. While it provides fast screening using acetone analysis, it is designed as a preliminary assessment tool rather than a clinical diagnostic device. The trade-off lies between comfort + affordability versus clinical precision, making the breath device ideal for frequent screening, while confirmatory blood tests remain necessary for medical diagnosis.

1. Pain & User Experience

- **Breath Device:** Completely non-invasive, painless, straw-based.
- **BeatO / Glucometers:** Requires finger-prick, causes discomfort, especially for daily users.
- **Lab Tests:** Blood draw needed; inconvenient for daily monitoring.

2. Recurring Cost

- **Breath Device:** Only ₹0.5/day (straw cost) → ~₹183/year.
- **BeatO:** Strip cost ₹14–16 per test → ~₹5,000–6,000/year.
- **Other Glucometers:** ₹18–23 per strip → ~₹7,000–8,000/year.
- **Labs:** ₹2,000–8,000/year depending on test frequency.
- **CGM:** Very high recurring cost (multiple 14-day sensors).

Trade-off of the Proposed Architecture

3. Daily Screening Feasibility

- **Breath Device:** Designed for quick 30–60 sec daily checks.
- **BeatO/Glucometers:** Possible but expensive and painful daily.
- **Lab Tests:** Not suitable for daily use.
- **CGM:** Excellent for continuous monitoring but extremely costly.

4. Portability

- **Breath Device:** Small, battery-powered, wireless.
- **BeatO:** Smartphone-dependent, requires strips & lancets.
- **Labs:** Not portable.
- **CGM:** Portable but high maintenance and cost.

5. Waste Generation

- **Breath Device:** Only biodegradable straw waste.
- **BeatO/Glucometers:** Lancets + strips → sharp waste & plastic waste.
- **Labs:** Biomedical waste.

Comparison:

METHOD / DEVICE	ONE-TIME COST (₹)	RECURRING COST PER YEAR (₹)	EXTRA YEARLY COST VS OUR BREATH DEVICE (₹)	NOTES
Our Breath Analyzer	₹4,000	₹183	—	Non-invasive, uses paper straw ₹0.5/test; 1 test/day = ₹183/year; screening device.
BeatO CURV Glucometer	₹499 – ₹1,279	₹5,292 – ₹5,840	₹5,109 – ₹5,657	Requires strips ₹14.5–₹16 each; Android Type-C only; finger-prick and lancets required.
Home Glucometer (General Brands: OneTouch/Accu-Chek/etc.)	₹600 – ₹2,400	₹6,870 – ₹8,330	₹6,687 – ₹8,147	Strips ₹18–₹23 each + lancets; painful; sharps waste.
Private Lab Testing (Monthly FBS + PPBS + HbA1c x3 per year)	—	₹2,700 – ₹8,100	₹2,517 – ₹7,917	Accurate but time-consuming; not suitable for daily use.
Government Hospital Testing	—	₹120 – ₹600	–₹63 to +₹417	Very low direct fees; long queues and travel; not practical for daily use.
Continuous Glucose Monitor (CGM: FreeStyle Libre, etc.)	₹2,000 – ₹4,000	≥₹25,000	≥₹24,800	Sensors ~₹5,000 per 14-day sensor; extremely high recurring cost; high accuracy.

Conclusion

- The project demonstrates that **non-invasive breath acetone measurement** can effectively support early diabetes screening. The device captures clear differences in breath responses between diabetic and non-diabetic individuals through controlled testing.
- The system integrates a **mixed sensor array, dual ADS1115 ADCs, and an ESP32 controller**, enabling stable, high-resolution data acquisition and reliable real-time processing. This ensures consistent performance across multiple trials and users.
- A **guided 30-second breath protocol**, combined with temperature–humidity compensation, improves accuracy by reducing environmental variations. This makes the device practical for real-world use rather than just laboratory conditions.
- The **Android application enhances usability** by managing calibration, guiding the user during testing, displaying results, and storing data automatically. This creates a smooth end-to-end workflow from breath sampling to result interpretation.
- Compared to conventional blood-based tests and devices like BeatO, the proposed system offers a **painless, hygienic, and extremely low-recurring-cost alternative**. Its portability and simplicity make it suitable for homes, clinics, and community screening camps.

Future Work

1.Integration of Machine Learning:

Future versions of the device can incorporate machine-learning models to improve classification accuracy and provide more reliable differentiation between healthy and diabetic breath profiles.

2.Improved Enclosure and Airflow Design:

A professionally designed 3D-printed enclosure with a dedicated airflow path and disposable mouthpieces can enhance sampling consistency, hygiene, and overall usability.

3.Cross-Platform App and BLE Support:

Adding Bluetooth Low Energy (BLE) and developing an iOS-compatible application will expand accessibility and reduce power consumption during wireless communication.

4.Enhanced Sensor Calibration & Miniaturisation:

Automated calibration routines, optimised power circuitry, and PCB-based integration can reduce sensor drift, improve stability, and make the device more compact and user-friendly.

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